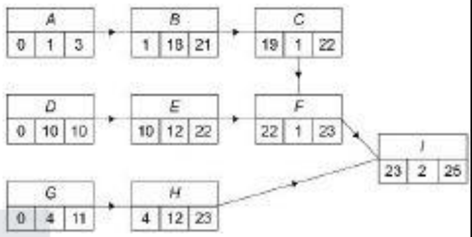
Mark schemes

Q1.

<table>
<tr><th></th><th>Marking Instructions</th><th>AO</th><th>Marks</th><th>Typical Solution</th></tr>
<tr><td>(a)
(i)</td><td>Correctly completes precedence table</td><td>AO1.1b</td><td>B1</td><td>D
C, E
G
F, H</td></tr>
<tr><td>(ii)</td><td>Correctly states earliest start times for all activities</td><td>AO1.1b</td><td>B1</td><td rowspan="2">
</td></tr>
<tr><td></td><td>Correctly states latest finish times for all activities</td><td>AO1.1b</td><td>B1</td></tr>
<tr><td>(b)
(i)</td><td>States the correct activity (label or in words) from their completed activity network
Only FT if their answer is A, D or G</td><td>AO1.1b</td><td>B1F</td><td>D, Chop vegetables</td></tr>
<tr><td></td><td>Identifies their activity as critical

OR

States that the float of their activity is zero</td><td>AO2.5</td><td>E1F</td><td>D is the first activity on the critical path</td></tr>
<tr><td>(ii)</td><td>States their correct time using their completed activity network

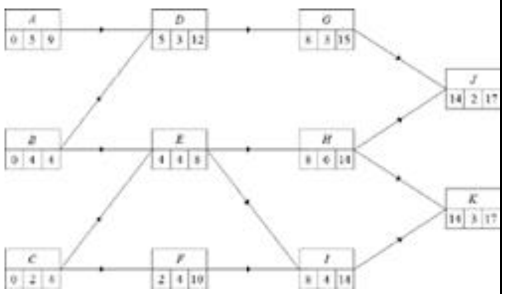
6:30 pm + their minimum completion time

Must see indication of pm OE</td><td>AO3.2a</td><td>B1F</td><td>6:55 pm</td></tr>
<tr><td>(c)
(i)</td><td>Completes precedence table 2 with correct durations

Condone correct durations in seconds</td><td>AO3.1b</td><td>B1</td><td><table><tr><th>Duration</th><th>Immediate Predecessors</th></tr><tr><td>10.25</td><td>−</td></tr><tr><td>8.25</td><td>J</td></tr><tr><td>0.5</td><td>K</td></tr></table></td></tr>
<tr><td>(ii)</td><td>Uses their minimum completion time and 19 in consistent units to find the float in the second project</td><td>AO1.1a</td><td>M1</td><td>25 − 19 = 6 minutes</td></tr>
<tr><td></td><td>States the correct time CAO

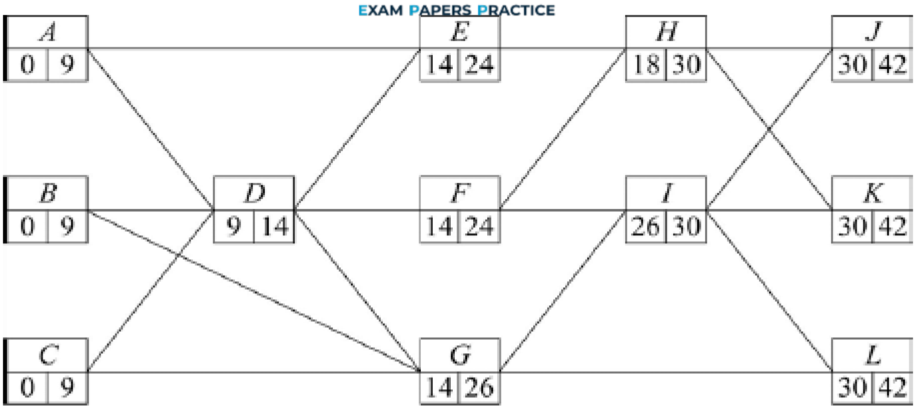
Condone omission of pm</td><td>AO1.1b</td><td>A1</td><td>6:36 pm</td></tr>
<tr><td></td><td colspan="4" align="right">Total 9 marks</td></tr>
</table>

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the correct earliest start time for each activity	AO1.1b	B1	
	Finds the correct latest finish time for each activity	AO1.1b	B1	
(b)	Evaluates the effects of reducing the duration of one of the activities on the project completion time	AO3.1b	M1	Reducing activity *E* duration to 1 hour reduces the project completion time to 14 hours, whereas all other activities reduce the project completion time to 15 hours or more
	Compares the effects of reducing the duration of each of the critical activities on the project completion time	AO3.1b	M1	
	Correctly deduces the activity which should have its duration reduced to one hour, from correct reasoning	AO2.2a	R1	Activity *E* clearly identified
(c)	Correctly identifies a limitation in the context of the project	AO3.5b	B1	Time between one activity ending and the next activity starting is not taken into account, as workers may need to travel to a different location
	Explains how the limitation that has been identified affects project in time or monetary terms	AO2.4	E1	Not taking into account time. The travelling time will cause subsequent activities to be delayed, increasing the project completion time
				Total 7 marks

Q3.

(a)



Forward pass, correct at D, E, F, G

M1

All correct

A1

Backward pass, correct at H, I, G *ft*

M1

All correct

A1

4

(b) $C\ D\ G\ I\ J$ only

B1

1

(c) 6

Their (latest – earliest – 4)

B1ft

1

(d) H delayed by 4

E1

K delayed by 5

B1

New time 51

51 scores 3/3

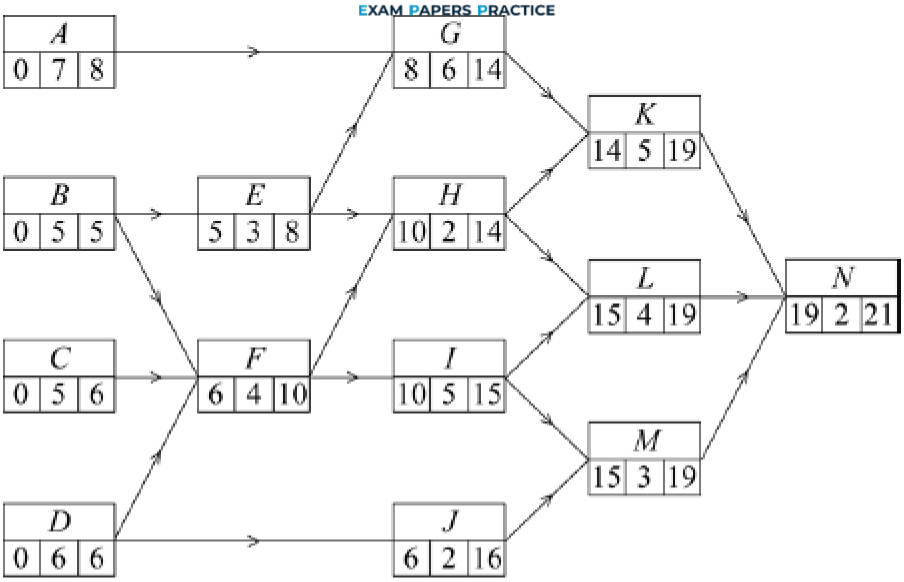
B1

3

[9]

Q4.

(a)



Forward pass

condone one slip (follow through) M1

all correct A1

Backward pass

condone one slip (follow through) M1

all correct A1

4

(b) Critical paths

$B\ E\ G\ K\ N$

first path correct M1

$D\ F\ I\ L\ N$

second path and no others A1

Minimum completion time is 21 days B1

3

(c) Cascade diagram

may be in blocks or bars (see examples)

One of ‘their’ CPs correct

ft their CP

M1

$B, D, E, F, G, I, K, L, N$

these activities correct

A1

A, C, H, J, M

3 of these with correct start and duration

M1

3 correct with correct slack indicated

A1

all 5 correct with correct slack

A1

		Slack
A	0 − 7	7 − 8
C	0 − 5	5 − 6
H	10 − 12	12 − 14
J	6 − 8	8 − 16
M	15 − 18	18 − 19

5

(d) (Max value of x is) 10

considering $J_{latest} - J_{earliest}$

M1

$\Rightarrow x \leq 10$

(condone $x < 11$ for SC2)
NMS $x \leq 10$ award M1 A1

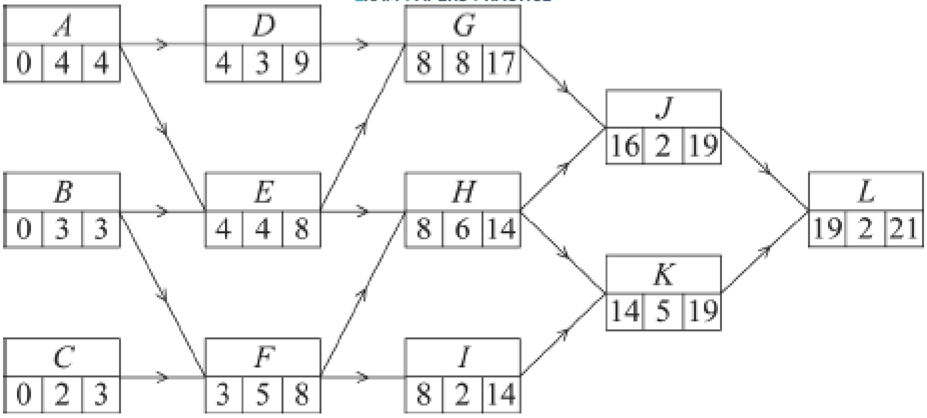
A1 cao

2

[14]

Q5.

(a)



Earliest start times

one slip follow through

M1

all correct

A1

Latest finish times

one slip follow through

M1

all correct

A1

4

(b) Critical paths are $AEHKL$ and $BFHKL$

one correct

M1

both correct and no extras

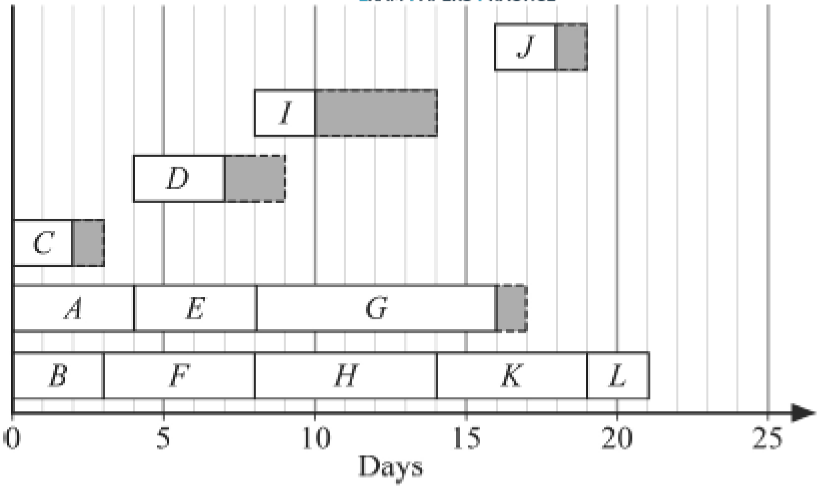
A1

Minimum completion time = 21 days

B1

3

(c)



A (0 → 4)
B (0 → 3)
C (0 → 2 → 3)
D (4 → 7 → 9)
E (4 → 8)
F (3 → 8)
G (8 → 16 → 17)
H (8 → 14)
I (8 → 10 → 14)
J (16 → 18 → 19)
K (14 → 19)
L (19 → 21)

A, B, E, F, H, K, L correct

B1

C, D, G, I, J
(4 with correct start and duration)

M1

All 5 correct with correct slack indicated

A1

3

(d) (i) K now starts day 17
or "delayed" b 3 days if 14 in network

B1

L now starts day 22
or "delayed" b 3 days if 19 in network

B1

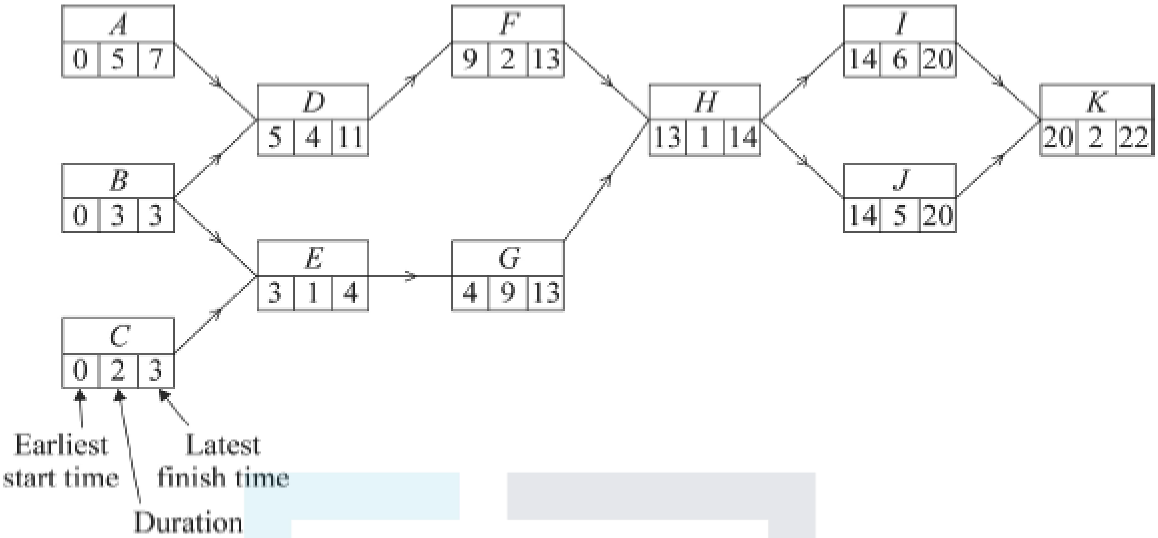
2

(ii) Overall delay 3 days

B1

1

Q6.



(a) Network attempted (3 more activities)
SCA

M1

Up to 2 slips (boxes or connections)

A1

Correct network
Condone missing arrows if sequence is clear

A1

3

(b) (i) Forward pass
up to 1 slip ft

M1

Correct

A1

2

(ii) Backward pass
up to 1 slip ft

M1

Correct

A1

2

(c) Minimum completion time 22 days

Must be stated – not simply in K box

B1

Critical path $B\ E\ G\ H\ I\ K$

and no others

B1

2

(d) (i) New start time for H is 15 days

For H, their (F earliest time 9) + (2 + 4)

M1

New start time for I is 16 days

both correct

A1

2

(ii) Minimum delay is 2 days

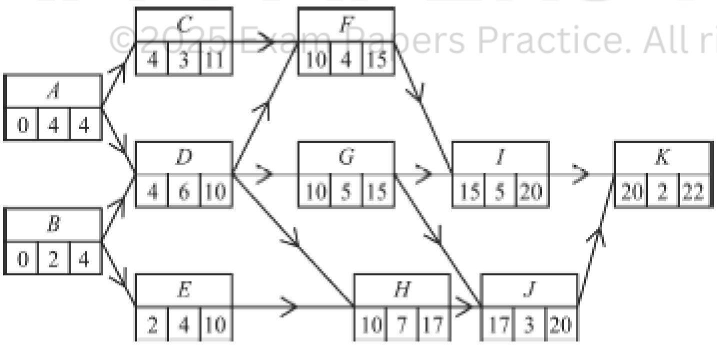
Condone new completion time 24 days

B1

1

[12]

Q7.



(a) Earliest start times

up to 2 errors ft

M1

All correct

A1

Latest finish times

up to 2 errors ft from K

M1

Correct

A1

4

(b) Critical paths $A\ D\ G\ I\ K$

$A\ D\ H\ J\ K$

withhold if extra path such as
$A\ D\ G\ J\ K$ *given*

B1B1

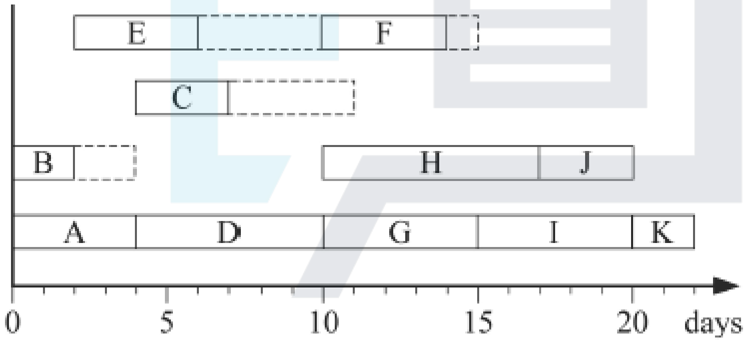
Minimum time for completion = 22 days

Must be stated – not just a value in box

B1

3

(c)



A, D, G, H, I, J, K correct

B1

B, C, E, F

2 correct with slack /float or 4 correct & no slack

B1

all correct with slack/float; withhold if slack not shown dotted etc

B1

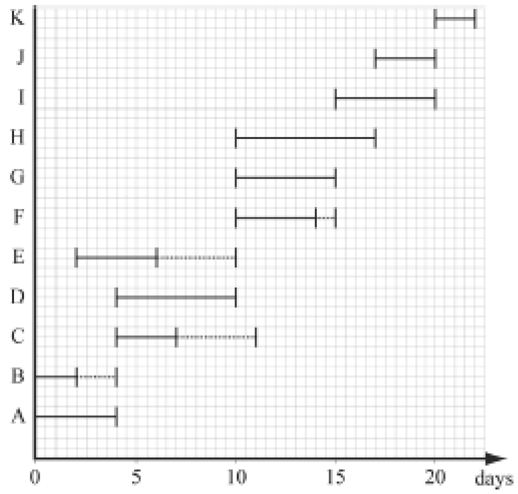
3

(d) F starts after 12 days at earliest
or F starts 2 days later
I is now unable to start until after 16 days
or starts 1 day later

E1

Minimum completion time now 23 days – one extra day etc

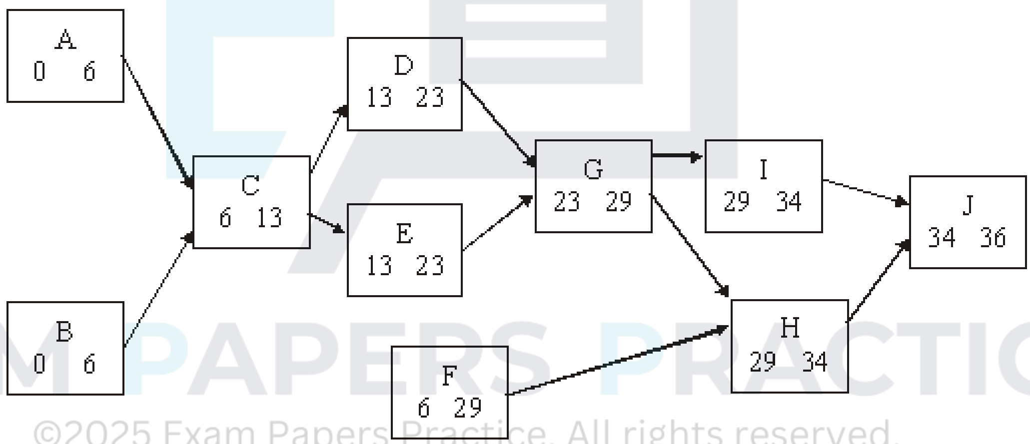
B1



2

[12]

Q8.



(a) Network

SCA

M1

–1 ee

A3

4

(b) Forward pass

M1

All correct

A1

2

(c) Backward pass

M1

All correct

A1

2

(d) (i) Project completion time 36 hours

B1ft

1

(ii) Critical path *BCEGHJ*

SCA

M1

All correct

A1

Earliest start + activity duration = latest finish time

E1

3

(e) (i) *I* now has new earliest time 29 + 3

Extra 3 hours on edge HI or new activity between H and I of duration 3

M1

= 32

A1

2

(ii) *I* now becomes critical and increases *J* earliest start time to 35

M1

New completion time is 37 hours

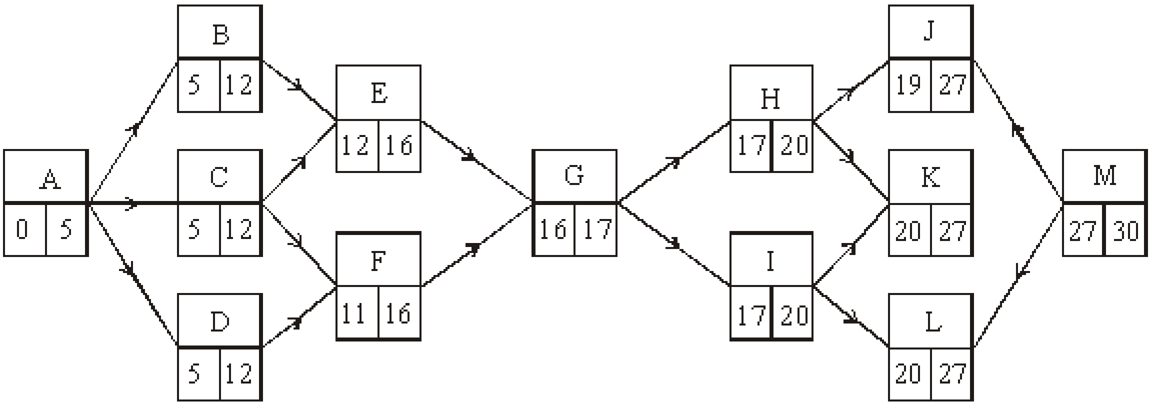
A1

2

[16]

Q9.

(a)



Forward pass

M1 A1 **2**

Back pass

M1 A1 **2**

(b) Critical path *ABEGILM*

B1 **1**

(c) *D*

B1 **1**

(d) (*C* now 9, so *E* starts at 14)

SC 32 scores ½

M1

Overrun 2 days

A1 **2**

(e) *F* start at 14

SC 35 scores 2/3

M1A1

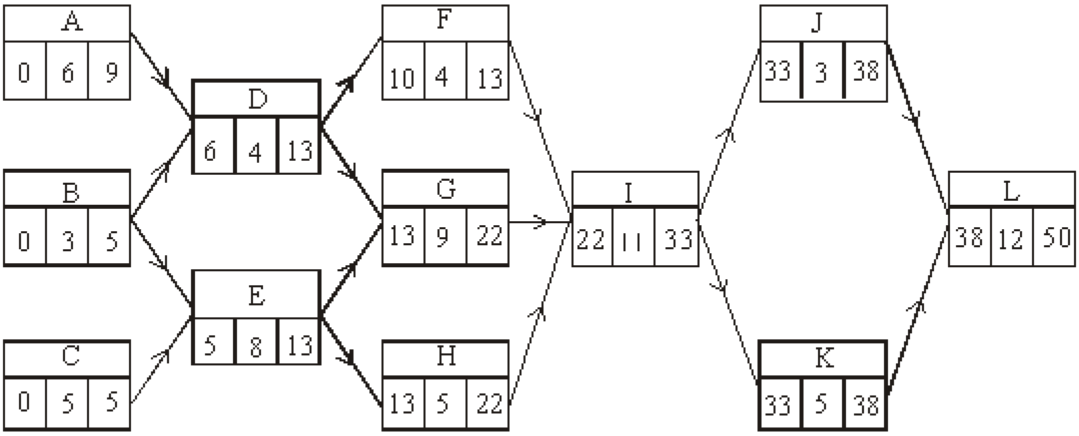
∴ overrun 5 days

A1 **3**

[11]

Q10.

(a)



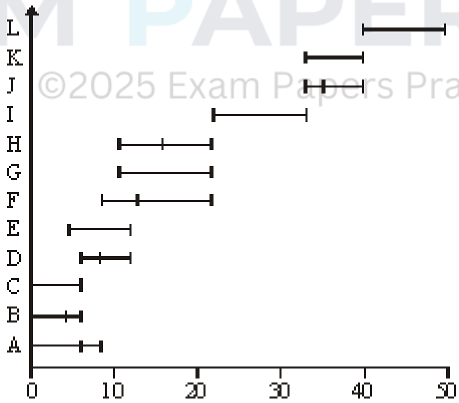
forward pass M1A1 **2**

back pass M1A1 **2**

(b) $C\,E\,G\,I\,K\,L$ B1 **1**

(c) F B1 **1**

(d)



Gantt diagram M1

floats included B1

correct, excluding floats A1 **3**

(e) (i) *D* 5 days $\Rightarrow$ *G* back 2 days

E1

∴ *J* starts at 35

OE

M1

∴ *L* starts at 43

∴ finish at 55

A1

3

(ii) *A D G I J L*

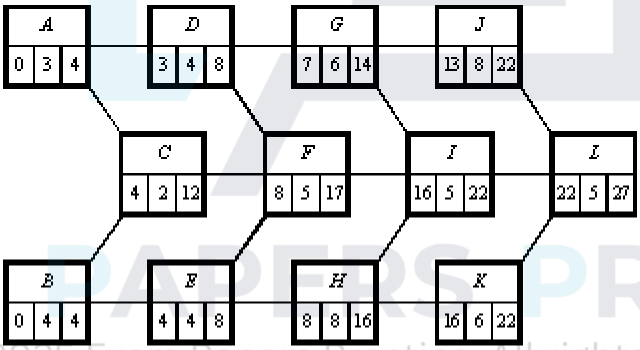
B1

1

[13]

Q11.

(a) (i)



Start times

M1 A1

2

Finish time

M1 A1

2

(b) *B E H K L* (= 27)

B1

1

(c) (i) New min time = 24

M1

Path *B E H I L*

A1

2

(ii) Leave *A, C, F* (non crit)

B1

Speed *B* by 2 (crit)

B1

J by 1 (becomes crit)

B1

K by 1 (non crit)
Reason

B1B1

5

(iii) Cost = £79 000

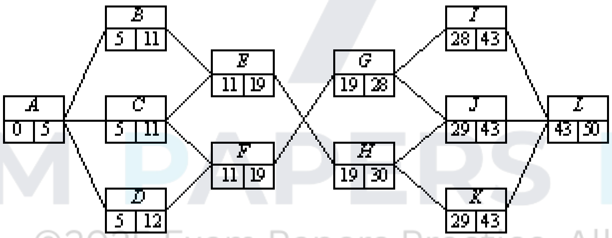
B1

1

[13]

Q12.

(a) (i)



M1

start times

A1

2

(ii) *finish times*

M1

A1

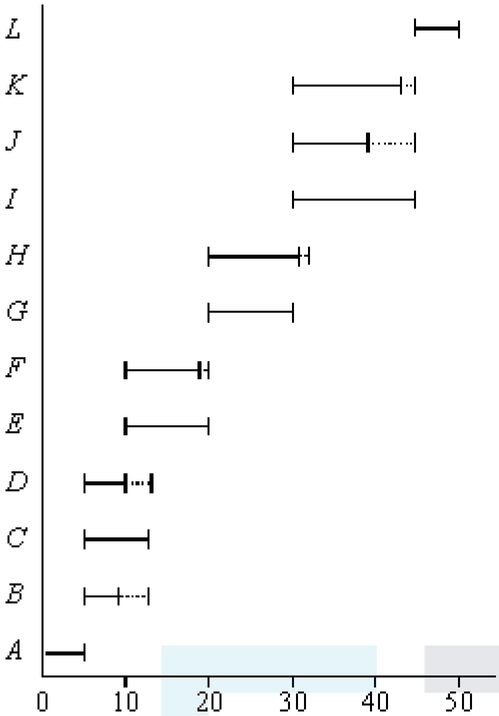
2

(b) Critical path *A C E G I L*

B1

1

(c)



M1

– *1 e.e.*

A2F

3

(d) From network
Problem with *B, C, D* and *I, J, K*

M1

Minimum extra *B, J*

A1

= 50 + 1 + 6 = 57

B1

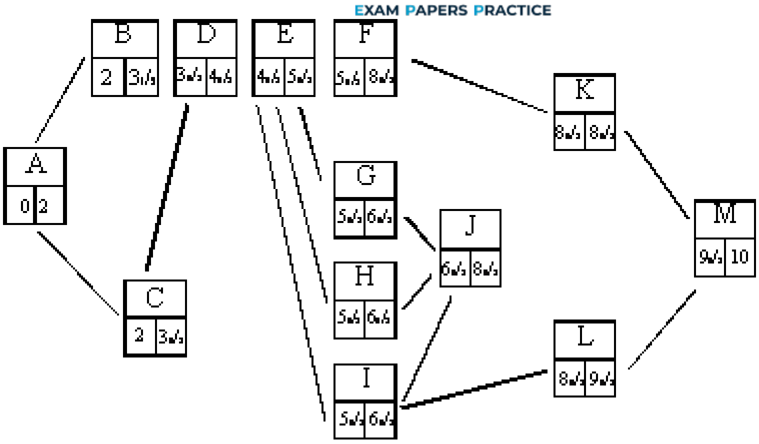
3

$$\begin{pmatrix} B + D - C = 1 \\ J + K - I = 6 \end{pmatrix}$$

[11]

Q13.

(a)



SCA

M1

– 1 each independent error

A3

4

(b) As diagram M1 3

–1 EE

A2

(c) As diagram

M1

3

– 1 EE

A2

(d) Critical A, C, D, E, H, I, J, K, M

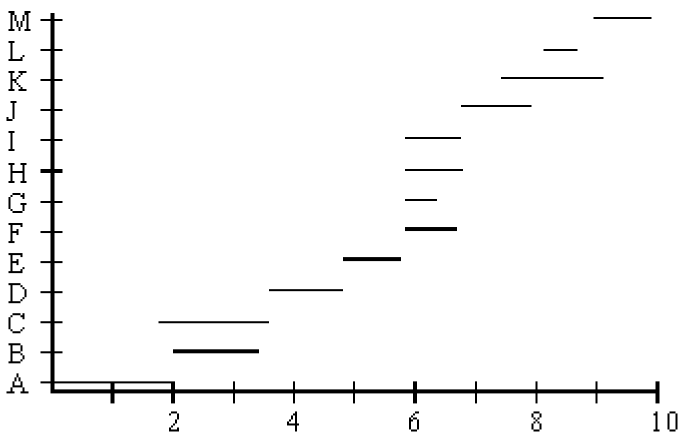
B1

10 days

B1

2

(e)



SCA

M1

–1 EE

A2f

3

(f) Electrician B, G – $\frac{3}{4}$ day

M1

Plumber C, (I) – $\frac{3}{8}$ day

A1

Joiner (H), J – $\frac{2}{5}$ day

M1

Use joiner – electric not critical

A1

New completion time $9\frac{3}{5}$

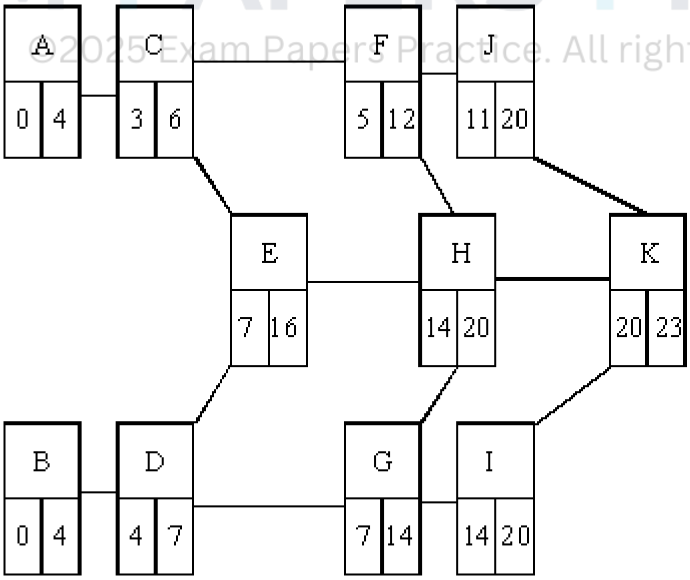
cao

5

[20]

Q14.

(a)



–1 each independent error

M1A3

4

(b) As diagram

M1A2 3

(c) As diagram

M1A1 3

(d) BDGIK

B1

23 weeks

B1B2 2

(e) (i) Leave E, float time 4

E1

Hurry I by 2 on C.P.

E1

Hurry J by 1 more on C.P

E1

No more because of H

E1 4

Shortest time 21 weeks

B1 1

(ii) 4000 × 2 + 3000

M1

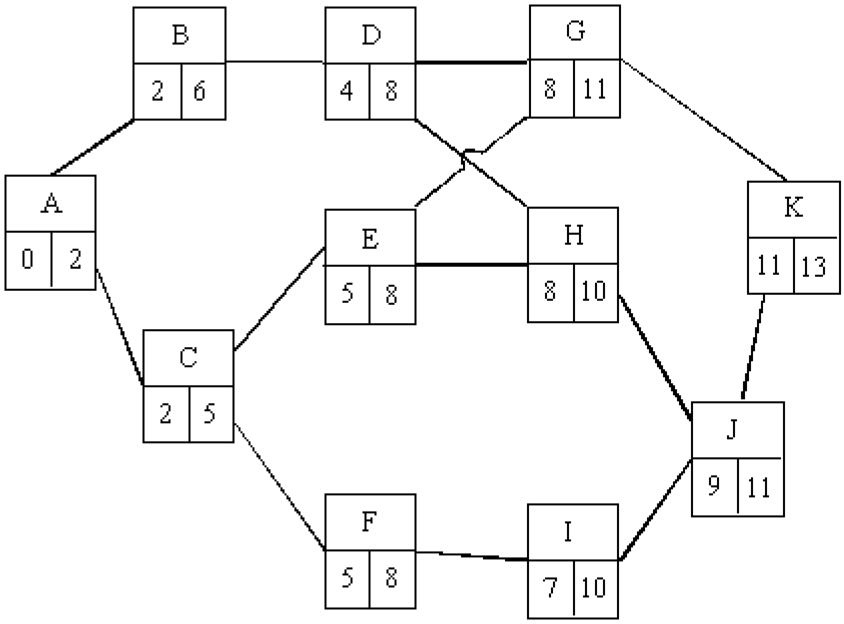
= 11000

A1 2

[19]

Q15.

(a)



– 1 each independent error

M1A3 **4**

(b) As diagram

A2 – 1 EE

M1A2 **3**

(c) As diagram

A2 – 1 EE

A1A2 **3**

(d) ACEGK
13 days

B1 **2**

Allow marked on diagram

B1

(e) overrun B and D

B1B1 **2**

(f) (i) 14 days

B1 **2**

K held up 1 day

B1

(ii) ACEGK
BDFIJH

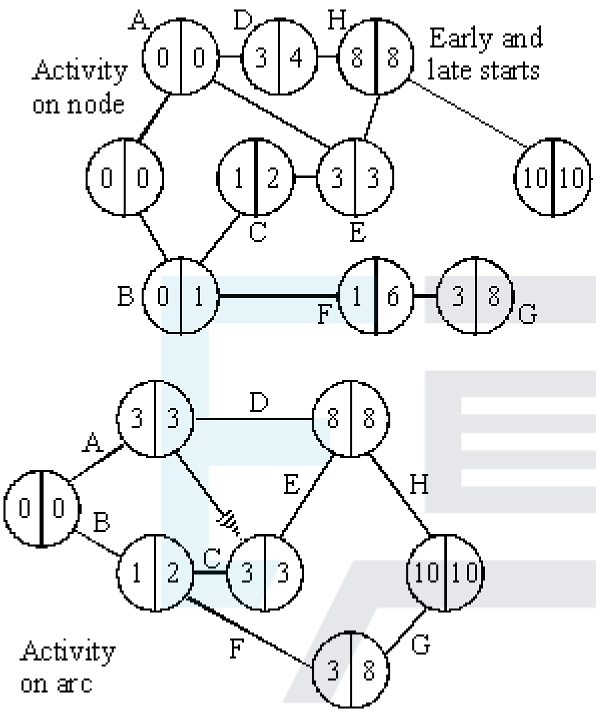
B1

1

[17]

Q16.

(a) + (b)



time:- 10

critical activities:- A, E, H

M1

Sca

A1

single start and finish

A3

(- 1 each error)

M1

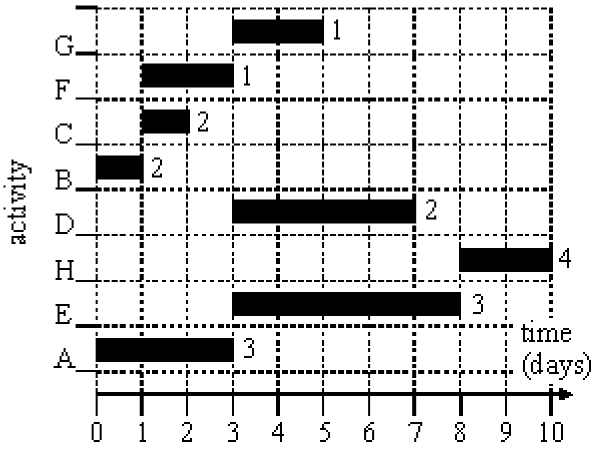
forward pass

A1 M1

backward pass

A1 B1 A1

(c)



($^{-1}$ each error)

M1 A2

(d) F – 2; G – 8; D – 4

B2 ($^{-1}$)

[16]

Examiner reports

Q1.

Nearly all students correctly completed the precedence table. Just under two thirds of students completed the activity network with no errors, with the most common error being simple numerical slips that were then carried forward through the network. Some students had values for the latest finish time lower than the earliest start time.

Many students correctly identified the first activity on the critical path and explained why this activity could not be delayed. A minority of students did not find the time at which the making of the meal would need to commence, instead just stating the minimum completion time of their activity network.

Part (c) was answered well, with the majority of students scoring 1 mark on (c)(i) and at least 1 mark on (c)(ii). The most common mistake was for students not to relate the minimum completion time of the second project to the time that the second project should start.

Q3.

The first three parts of this question proved to be a good source of marks for all students with many scoring full marks. There were problems in part (d), where students failed to appreciate the effect of BOTH H and K overrunning and the implications to the critical path.

Q4.

(a) Almost every student calculated the earliest start times and latest finish times correctly.

(b) At least one critical path was found correctly by almost everyone and most students found the two critical paths.

(c) Greater care should be taken when drawing diagrams of this type, preferably ***using a ruler***. The most common error in the cascade diagram was the omission of slack for the non-critical activities; others tried to show activities starting as late as possible. Some tried to insert activity J as two sections, so occasionally the diagram resembled a histogram.

(d) Those students who made errors in their earliest start times and latest finish times for J were given credit for writing down an inequality based on their values. Quite a large number of students did not realise the need to use the difference between the latest finish time and earliest start time for J only and so had a totally incorrect inequality.

Q5.

This proved to be a very high scoring opening question for most candidates.

Part (a) The earliest start times were usually correct, but the latest finish times for some activities were sometimes calculated incorrectly.

Part (b) Most candidates found at least one correct critical path, but sometimes spoiled their answers by adding a third path that was not a critical path. Some candidates listed all the critical activities instead of giving the critical paths and scored no marks for this.

Part (c) Most cascade diagrams indicated the critical activities correctly. Quite a lot of candidates failed to show the correct slack for activities C, D, G , I or J. Others had slack overlapping another activity and this was not acceptable.

Part (d) The last part of the question was answered well by most candidates, although some failed to mention the new earliest starting **day** for either K or L.

Q6.

This proved to be a very good opening question for all candidates. The earliest start times were usually correct but the latest finish times for D, B and A were sometimes calculated incorrectly. Most candidates found the correct critical path, but quite a few forgot to state the minimum completion time. Those with a correct activity diagram were usually able to find the new start times for H and I after the delay. A few thought that the minimum delay was 1 day and others stated that the delay was 24 days, when in fact this was the new completion time.

Q7.

This proved to be a very good opening question for all candidates. The earliest start times and latest finish times were usually correct. Some candidates only gave one critical path instead of two. Quite a large number thought that $ADGJK$ was also a critical path and lost a mark. Despite showing a value of 22 on their insert, several candidates forgot to state the minimum time for completion. The cascade diagram caused few problems. Some candidates used blocks with critical activities linked together and other activities drawn in blocks above; others used a sequence of horizontal lines with the vertical axis labelled from A to K. The main reason for losing marks in part (c) was the ignoring of the slack or float. The explanation in part (c) was not always adequate: it was necessary to explain that F needed to start after 12 days at the earliest, with I and K being delayed by one day, and hence that the new minimum completion time was 23 days.

Q8.

This was a high scoring question for practically everyone. The activity network in part (a) was usually done correctly and candidates seemed well drilled in critical path analysis. Similarly, parts (b), (c) and (d) were answered correctly by even the weakest candidates. The explanation in part (d)(ii) was not always clear. Part (e) was meant to be a little more challenging, but only the weakest candidates failed to find the new earliest start time for activity I and the revised shortest completion time for the project.

Q9.

This question proved to be a good source of marks for all candidates. A significant number of candidates failed to answer parts (d) and (e) correctly. Correct solutions were sometimes obtained by candidates amending their figure by including the overruns and their implications.

Q10.

This question proved to be a good source of marks for all candidates, although a significant number failed to score full marks in part (d) due to the inaccuracies of their diagram or through drawing a resource histogram. Part (e) discriminated well between candidates. This was the first time that candidates were provided with a stage/state table for answering a dynamic programming question. The question gave candidates the opportunity to use a network diagram and still score full marks. A significant number of weaker candidates thought that, as there were only six possible orders, a complete enumeration was an acceptable method. This was not the case. About half of the candidates worked backwards through the network and about half worked forwards. Both methods were equally successful.

Q11.

The majority of candidates scored full marks on parts (a) and (b), but part (c) discriminated well amongst the candidates. Although candidates could successfully find the new minimum completion time, they did not easily apply this time to implications for critical/non-critical activities.

Q12.

Parts (a), (b) and (c) were answered well by many candidates but part (d) proved to be the hardest question on the paper. There were some mistakes made on the backward pass to find the latest finish time for the activities. The majority of candidates used graph paper to draw their Gantt diagram, a practice that should be encouraged. In part (d) those candidates who looked at the original network realised that with only two workers there was a problem with B, C, D and I, J, K and found the minimum extra time that two workers would need to complete the three activities and arrived at the correct solution.

Q13.

The question involved a routine application of critical path analysis. The candidates were well prepared and scored nearly all the available marks in the first five parts. It was pleasing to note that candidates worked their critical path on their network diagram, with the benefit of saving time for the later, more searching, parts of the question. Although part (f) was challenging, there were a number of correct solutions with candidates able to relate the new information back to the critical path together with its implications.

Q14.

Candidates were able to score the majority of marks on parts (a) to (d), but part (e) proved to be challenging. Candidates needed to consider whether the speeding up of an activity would speed up the completion of the whole project and, after an activity has been speeded up, what the implications on other activities would be.

Q15.

This was a standard critical path analysis question where candidates were expected to use their diagram in part (a) to help them with the later parts of the question. Unfortunately too many candidates chose to compute each critical time using the full algorithmic method rather than using their activity network. The numbers used in a critical path question will normally be easy enough to allow candidates to use their network as a working diagram.

Q16.

This question was generally very well answered, with many candidates obtaining at least 14 of the available 16 marks. Almost everyone obtained the correct activity network, and almost all of the forward passes were correct. There were some slips made in the backward pass, particularly for activity B. Most obtained the correct minimum duration and list of critical activities.

In part (c) the majority of candidates completed the cascade chart correctly, although a few showed all of the activities starting at time zero. Far fewer could obtain the new scheduled start times in part (d).